# Ravi Kumar

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## Professional Summary

B.Tech CSE (AI & ML) student with experience in Machine Learning, Deep Learning, and Generative AI. Skilled in building real-world AI systems using YOLO, NLP, and RAG pipelines. Proficient in Python, TypeScript, and full-stack development, with hands-on experience in NFC-based systems and LLM applications.

## Education

**Kalasalingam Academy of Research and Education** *2022 - 2026*
B.Tech in Computer Science and Engineering (AI & ML)
**CGPA:** 7.96

**BSEB – Intermediate (12th Grade)** *2020 – 2022*
**Percentage:** 76.2%

**BSEB – Secondary Education (10th Grade)** *2019 – 2020*
**Percentage:** 78%

## Technical Skills

- **Languages:** Python, Java
- **Frontend:** React.js, HTML, CSS
- **AI/ML:** Machine Learning, Deep Learning (CNN), Computer Vision
- **Generative AI:** RAG, Prompt Engineering, LLM Applications
- **Frameworks/Tools:** TensorFlow, PyTorch, OpenCV, LangChain
- **Database:** MySQL
- **Core Concepts:** Data Structures & Algorithms (Basic)

## Projects

**Context-Aware AI Chatbot using LangChain** *2026*
- Built a RAG-based chatbot for querying PDF and web data using FAISS, HuggingFace embeddings, and Groq API (LLaMA 3).
- Achieved 95%+ answer relevance through optimized retrieval and deployed with a Streamlit interface.

**Smart Campus Card System** *2026*
- Developed a smart campus card system using NFC for secure payments, attendance, and access control.
- Built a scalable TypeScript full-stack application with real-time transactions and user management.

**Smart Surveillance System using YOLO** *2025*
- Developed a real-time object detection system using YOLO to detect weapons with 85% accuracy and trigger alerts.
- Integrated Groq API (LLaMA 3) for context-aware AI-generated security alerts, improving decision-making.

## Experience

**AI Trainer (Freelance) – Outlier AI** *jan 2025 - june 2025*
- Trained and improved large language models by designing and optimizing prompts for better accuracy, relevance, and response quality.
- Performed prompt engineering and evaluation tasks, refining model outputs through iterative testing and feedback.
- Analyzed AI responses to identify errors, biases, and inconsistencies, enhancing overall model performance.

## Certifications

- Computer Vision in Microsoft Azure – Coursera *2024*
- BEC English Certification *2023*